

Open CPX Optical Engine with Integrated Laser

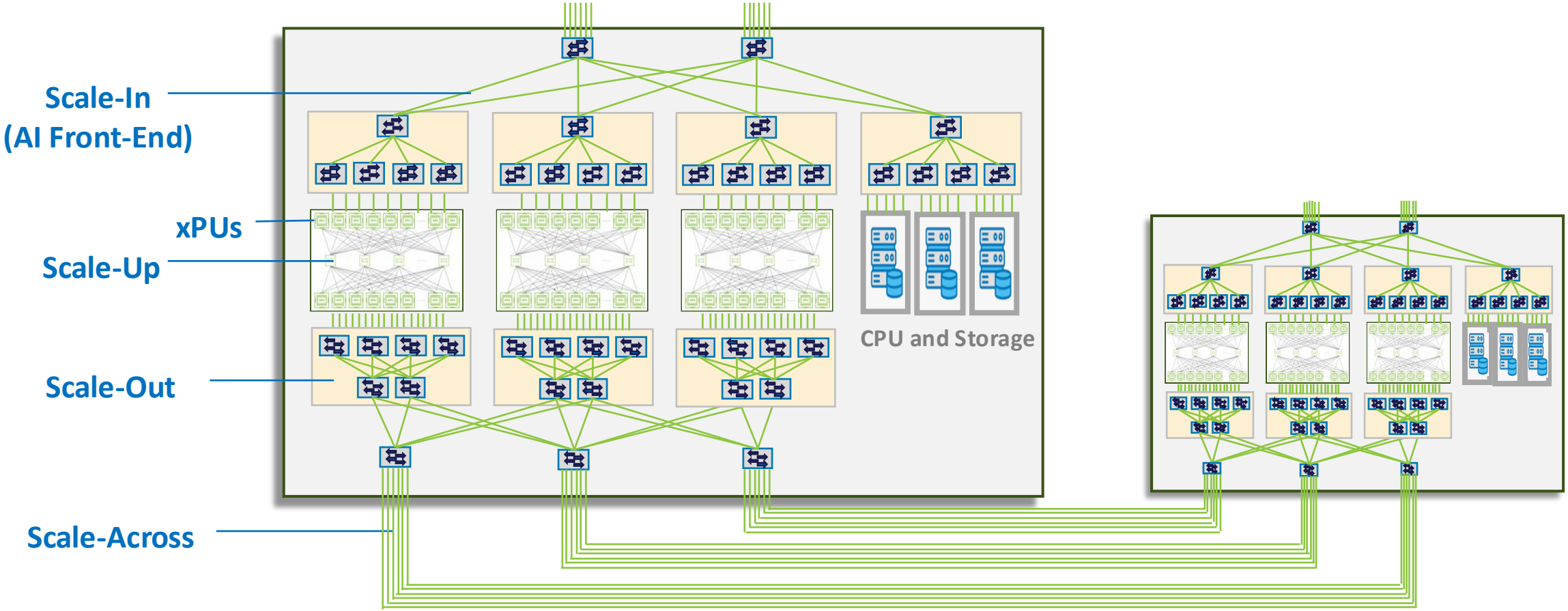
Rang-Chen (Ryan) Yu

VP Marketing, TeraHop



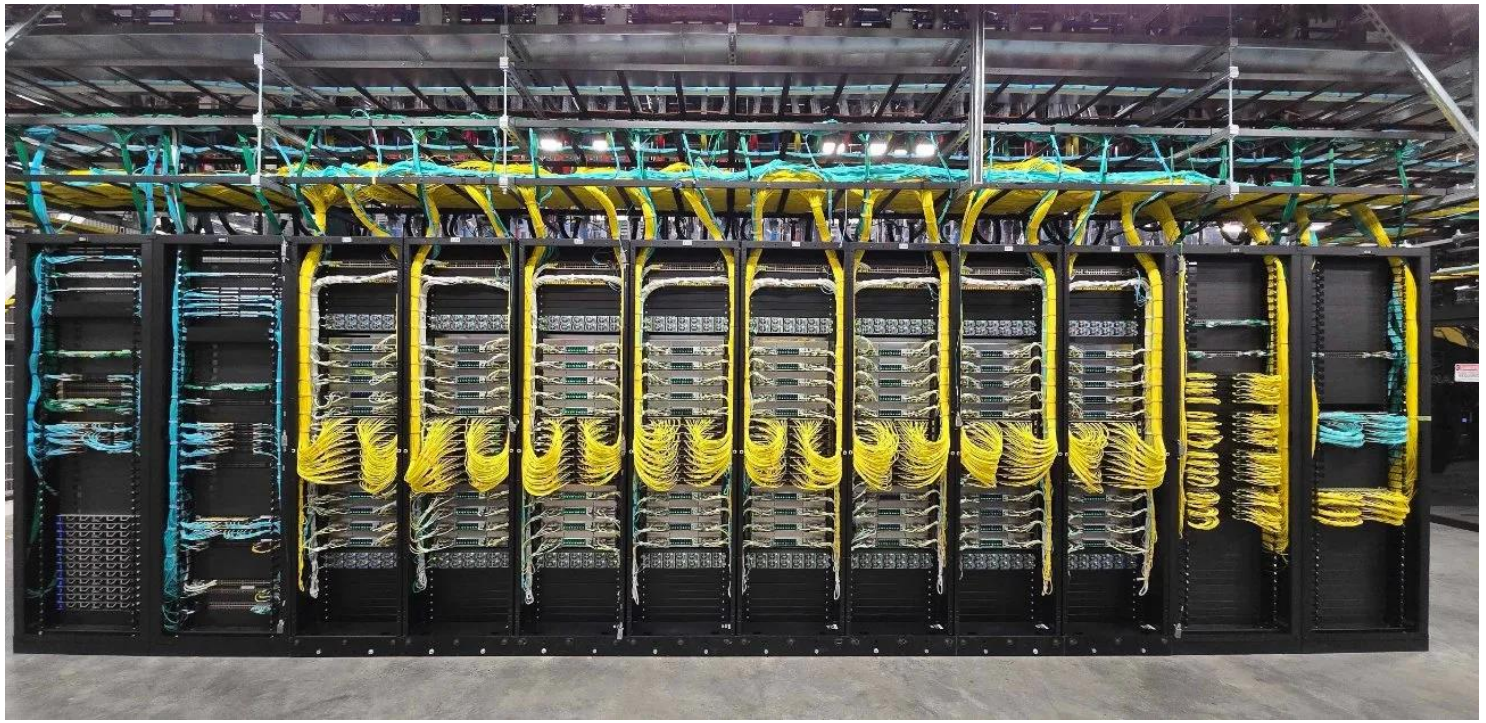
AI Scaling Networking

- AI Scaling requires complex networking with scale up, scale out, scale across and scale-in
- Optical connectivity solutions required with increasing challenges
  - Capacity, density, fiber, power,..



AI DC Optical Connectivity Challenges

- Optical Transceiver Capacity/density Challenges
  - AI scale up BW ~ 10x of scale out
    - e.g., NV VR Scale up~ 72 x 200G = 14.4Tbps vs. 1.6Tbps Connetx-9 Super NIC)
    - Transition from copper to optical: e.g. NVL72 copper → NVL576 with copper/optics



Source: Nvidia, NVIDIA Polyphe prototype, a fully functional GB200-based multirack NVL576 scale-up system

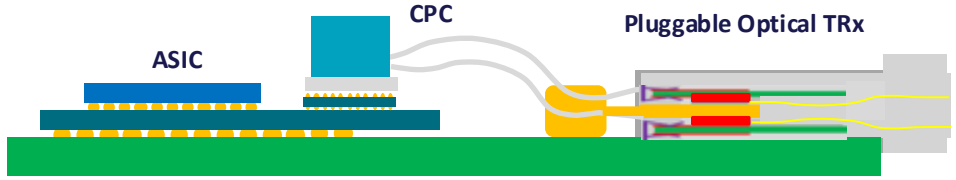
Optical Solutions Roadmap for AI Scaling





OSFP Pluggable Optical Transceiver

- 1.6T OSFP in full production
- IMDD: 1.6T → 3.2T ~ 2027-2028
- “Coherent Lite”: 1.6T-CL and 3.2T-CL for AI Scale Across

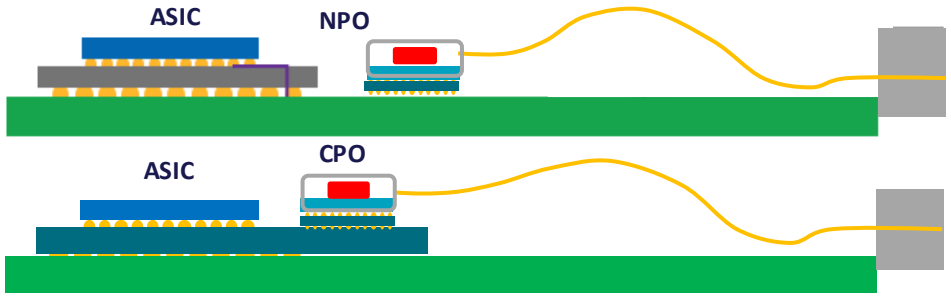


XPO OPTICS INNOVATION



12.8T XPO MSA

- XPO MSA announced at OFC 2026
- 1st demonstration of full XPO capabilities at OFC’2026
- Expect Production Deployment within 2027
- 25.6T XPO+ in early development



OPEN CPX MSA



6.4T Open CPX MSA

- Open CPX MSA Announced at OFC’2026
- Release 1.0 announced September 16, 2026
- Expect Production Deployment within 2027



“Slower and Wider” (SAW) for Scale Up

- Many “SAW” options: SiPho NRZ, uLED, uVCSEL
  - OCI MSA announced at OFC’2026
- PoC /demo deployment near term
- Volume deployment ~ ?

Open CPX MSA

OPEN CPX MSA

The Open CPX MSA founding members include Ciena, Coherent, Marvell, Molex, Samtec, and TeraHop (alphabet order).

The Open CPX MSA early Contributor members include Accton Technology, Intel, Nexthop Systems, Qualcomm and TE Connectivity Solutions.

- Open CPX MSA launched at OFC'2026
- 50+ members
- Release 1.0 on 9/17/2026

Open CPX MSA

Specification for

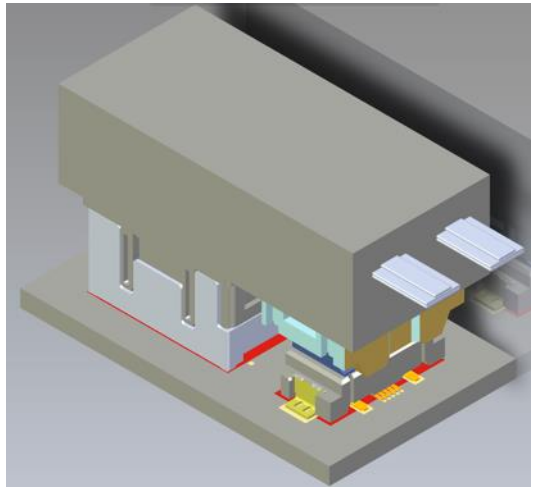
Co-packaged and Near-Package Connector and Optics Module

Rev 1.0

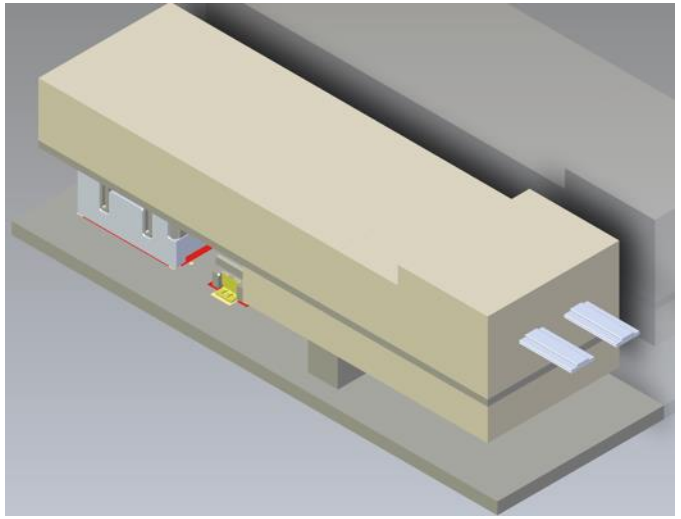
2026 September 16

Open CPX MSA Optical Engine Form Factors

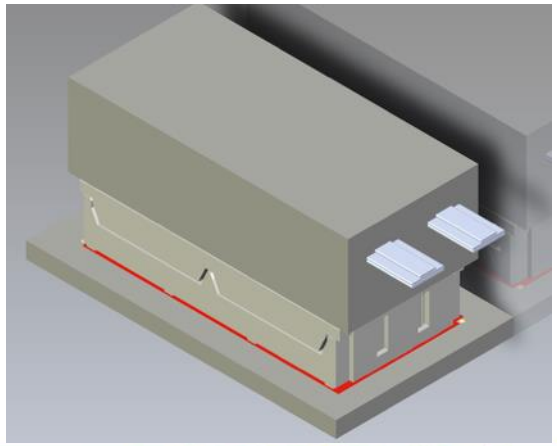
Type 1 Connector, ELM



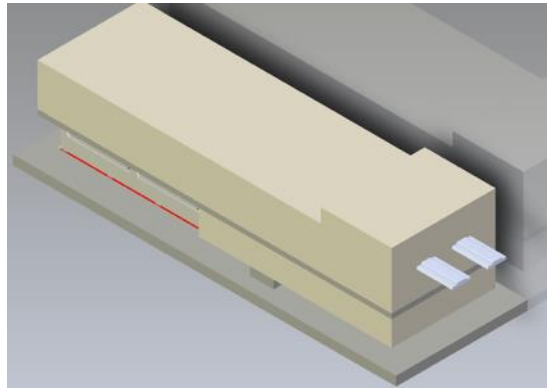
Type 1 Connector, ILM



Type 2 Connector, ELM

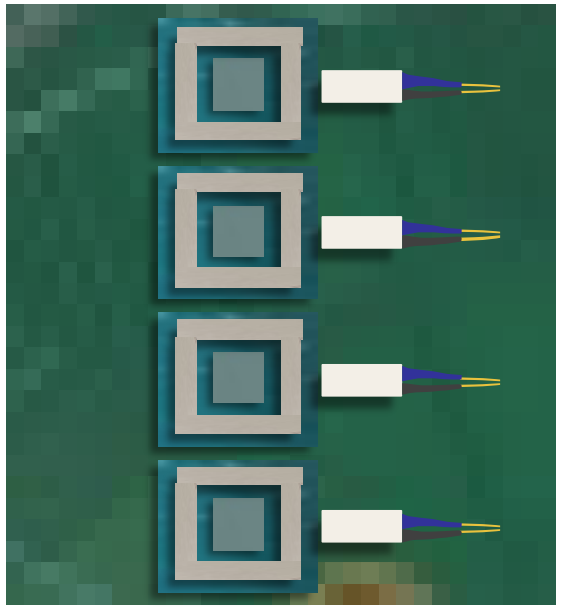


Type 2 Connector, ILM

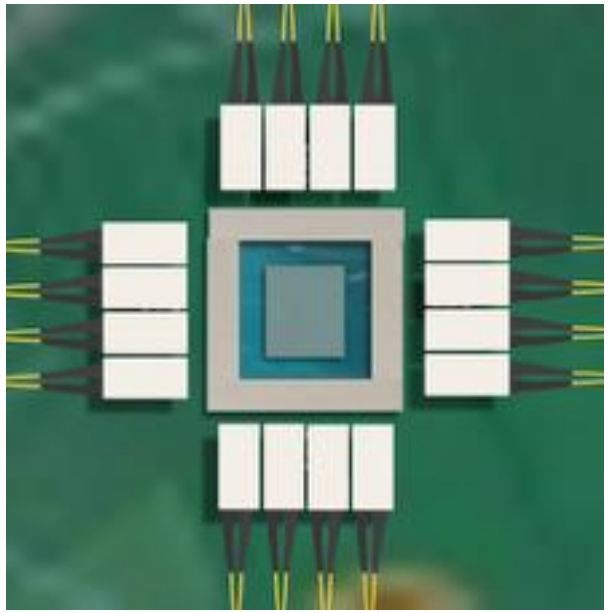


Use Case Examples of Open CPX Optics

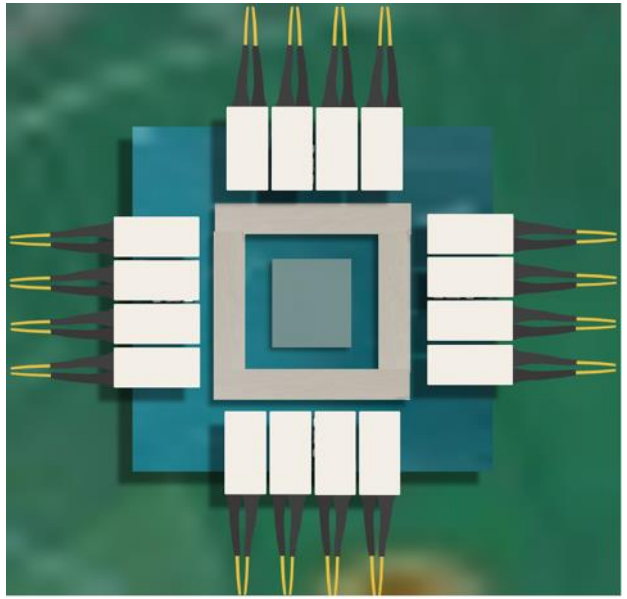
NPO for xPU Scale up



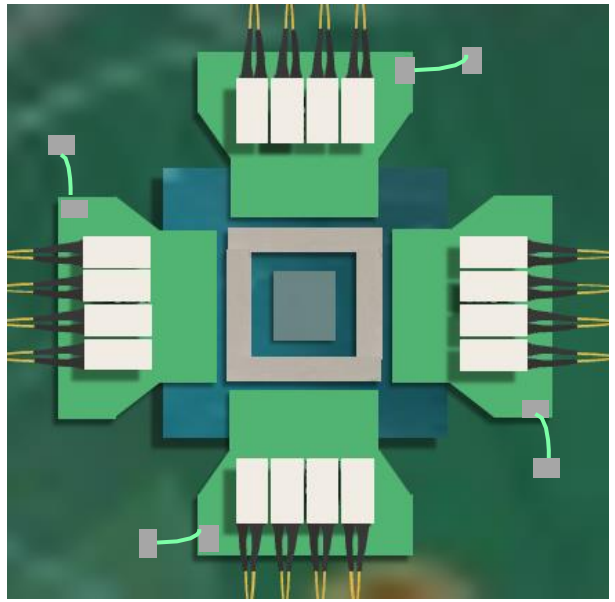
NPO for Switch Scale up/Out



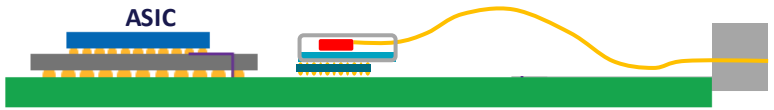
CPO for Switch Scale up/Out



IOA (Integrated Optical Attachment)



Open CPX NPO



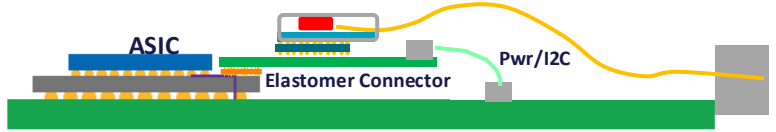
- Optical Engine on Host PCB
- Type 1 or Type 2 Connectors

Open CPX CPO



- Optical Engine on ASIC substrate
- Type 1 connector with high speed DP on ASIC substrate, power/I2C on host PCB

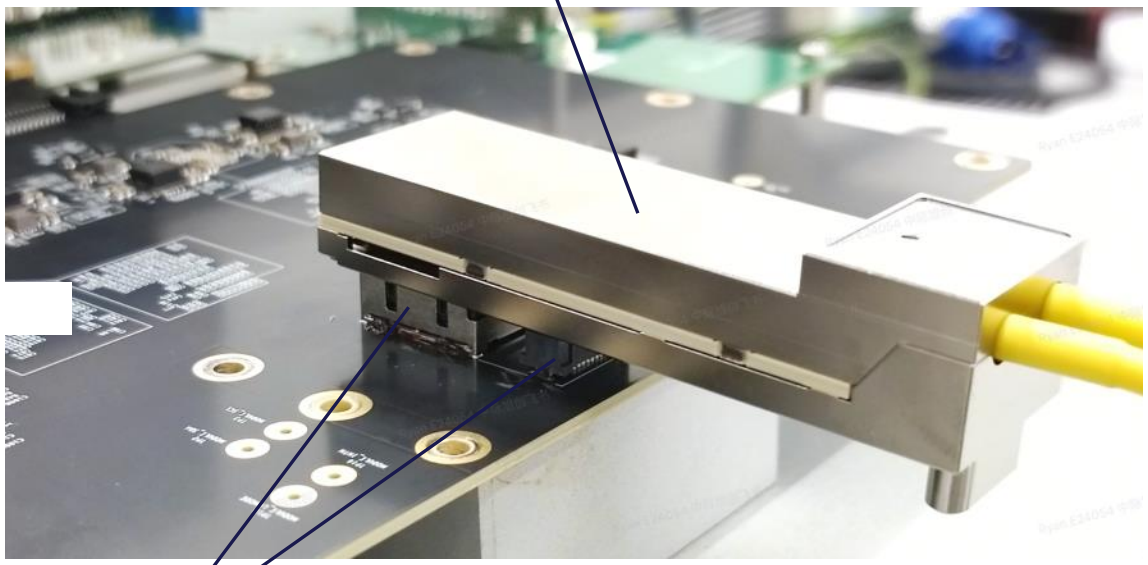
Open CPX IOA



- Optical engine on HDI board attach to ASIC substrate via elastomer high density, high speed low loss connector
- Type 1 or type 2 connectors

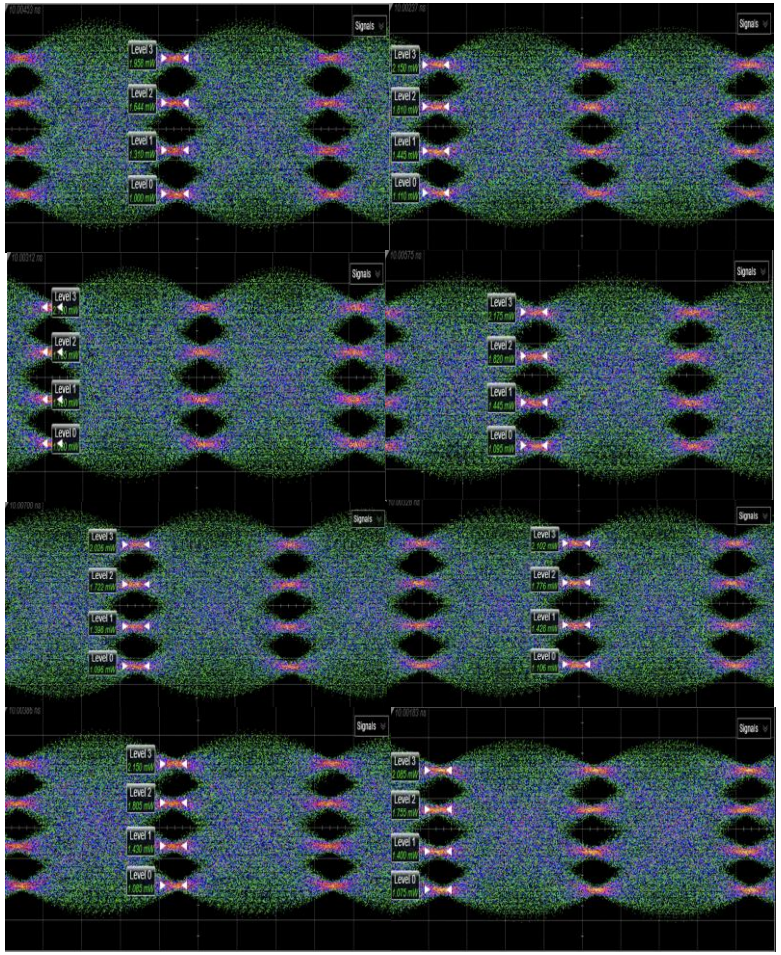
TeraHop “Diablo-1” 6.4T Open CPX Engine

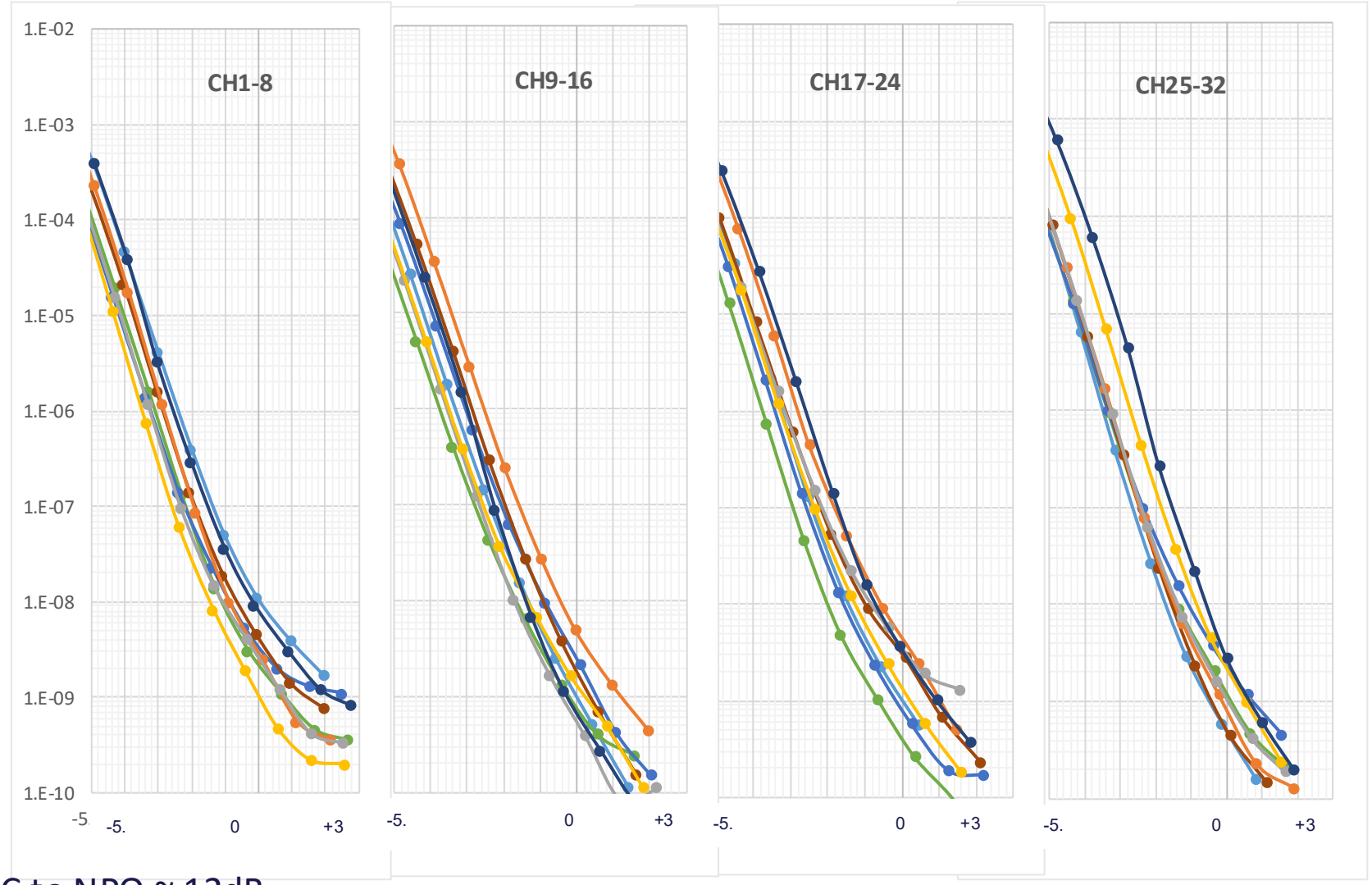
Silicon Photonics Optical Engine with Integrated laser Source



Open CPX Type 1 Connectors

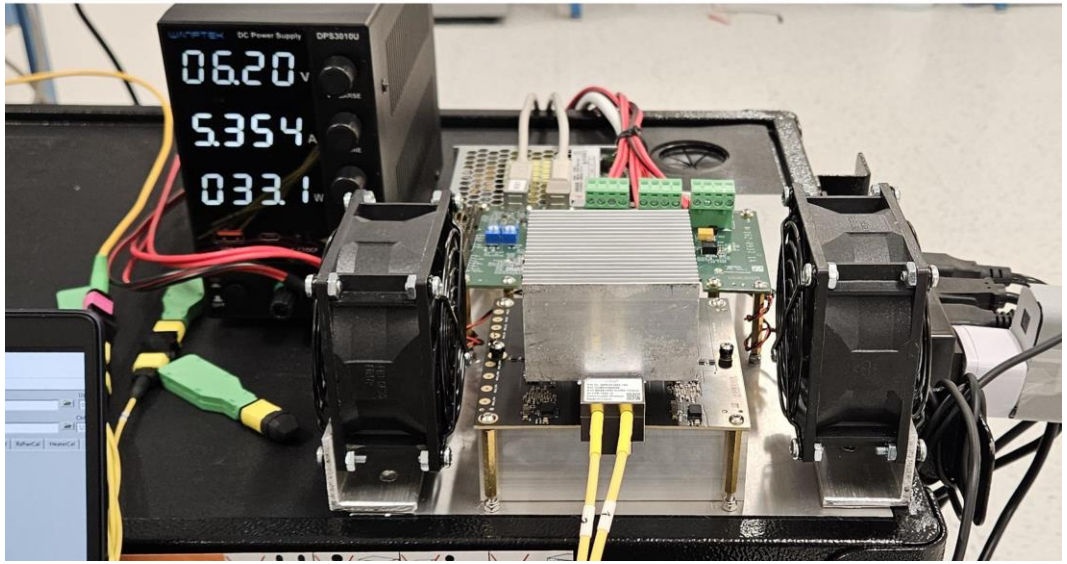
TeraHop “Diablo-1” 6.4T NPO Prelim. Test Data



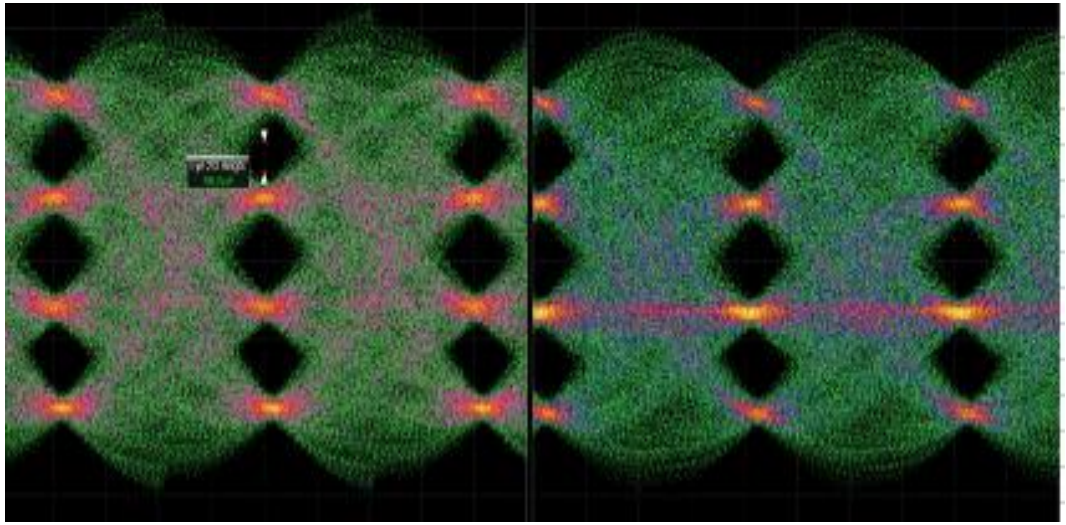


* Insertion loss between ASIC to NPO ~ 12dB

TeraHop “Diablo-1” Over 1km Fiber

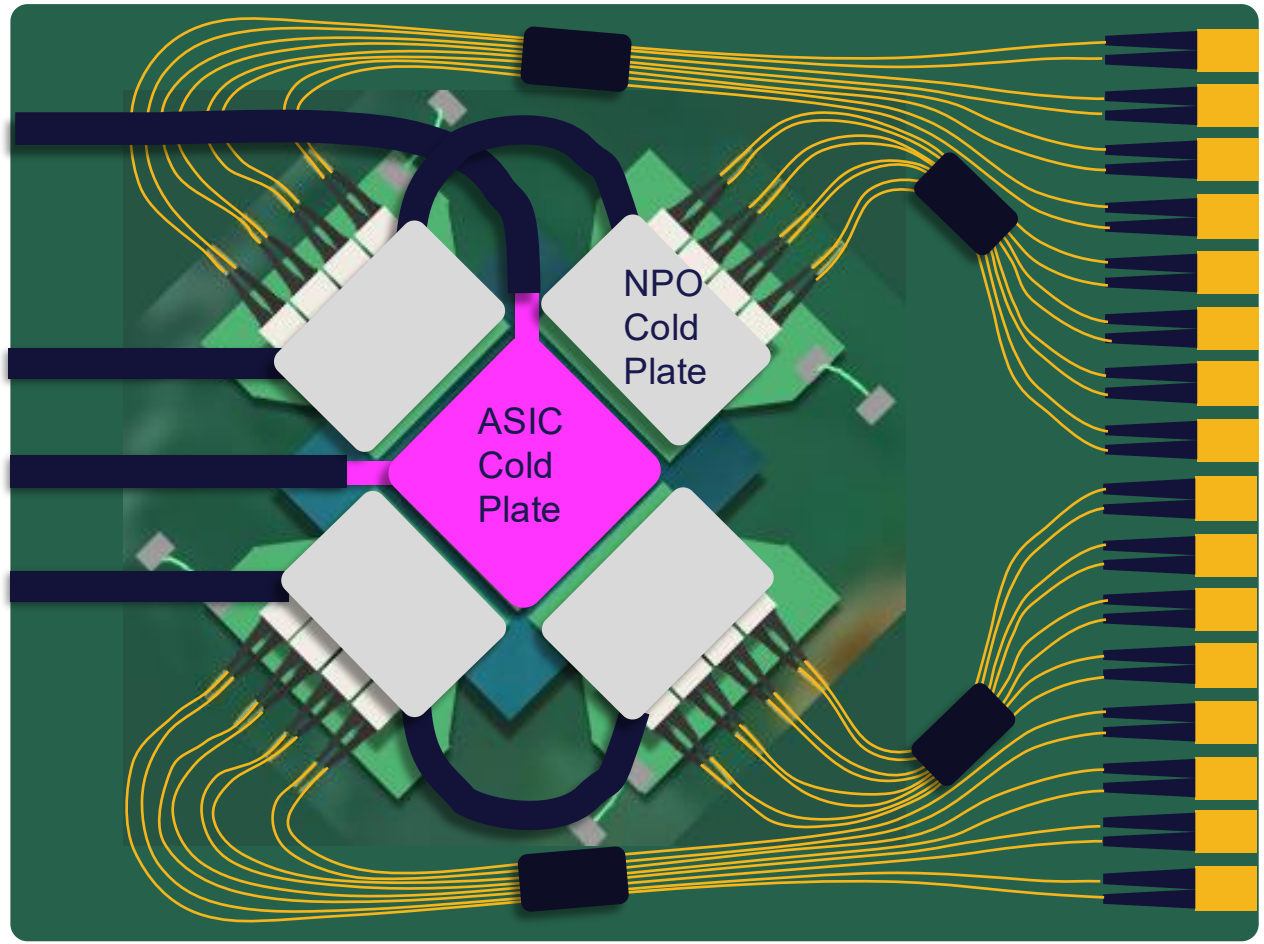






CH	EdStart	ErrCnt	BitCnt	_Ber	R_Lock	L_Lock
1		1.2879E+4	6.911775E+12	1.86334E-9		
2		4.255E+3	6.879475E+12	6.18506E-10		
3		8.579E+3	6.84675E+12	1.253E-9		
4		2.3518E+4	6.814663E+12	3.45109E-9		

“Diablo-1” 64.T NPO Key Attributes

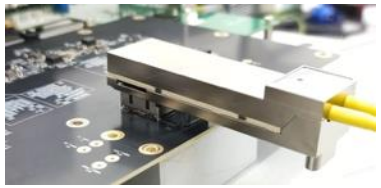


- Integrated laser with proven low FIT
- NPO close to, but separate from ASIC
  - Separate cold plate from ASIC
  - Critical components including LD at modest temperature
- Low power ~ 5.5pj/b, including internal laser
- Simple SMF pigtail termination to front panel
  - No extra connector loss and potential failure

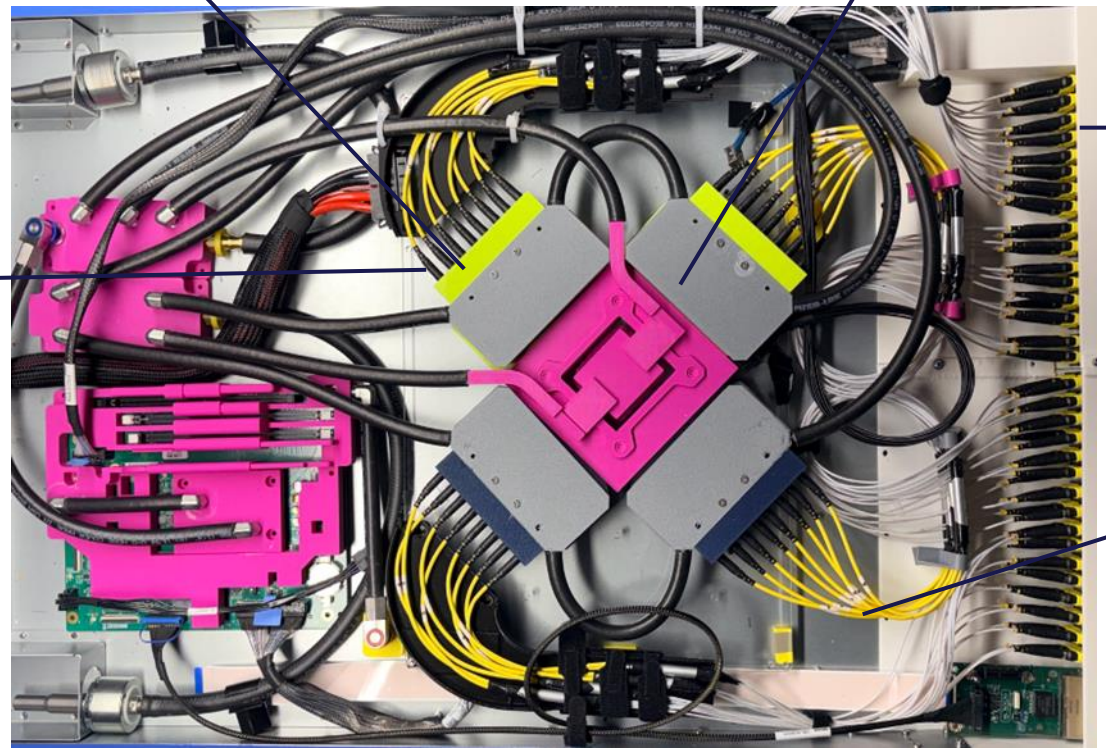
- ELSFP:
- lose >3dB power from extra coupling and connectors
- →requires > 2x CW LD optical power (200-300mW+)
  - lacking field reliability data
  - increase CW LD power dissipation by ~ 4x
- PM fiber adding additional power loss and cost
- Blind-mate connector adding loss and failure
- Higher cost, power and less reliable

Industry 1st 100T Switch with Open CPX Optics

- "Diablo-1" 6.4T NPO with integrated lasers,



- Integrated fiber pigtail, no failure-prone optical interface

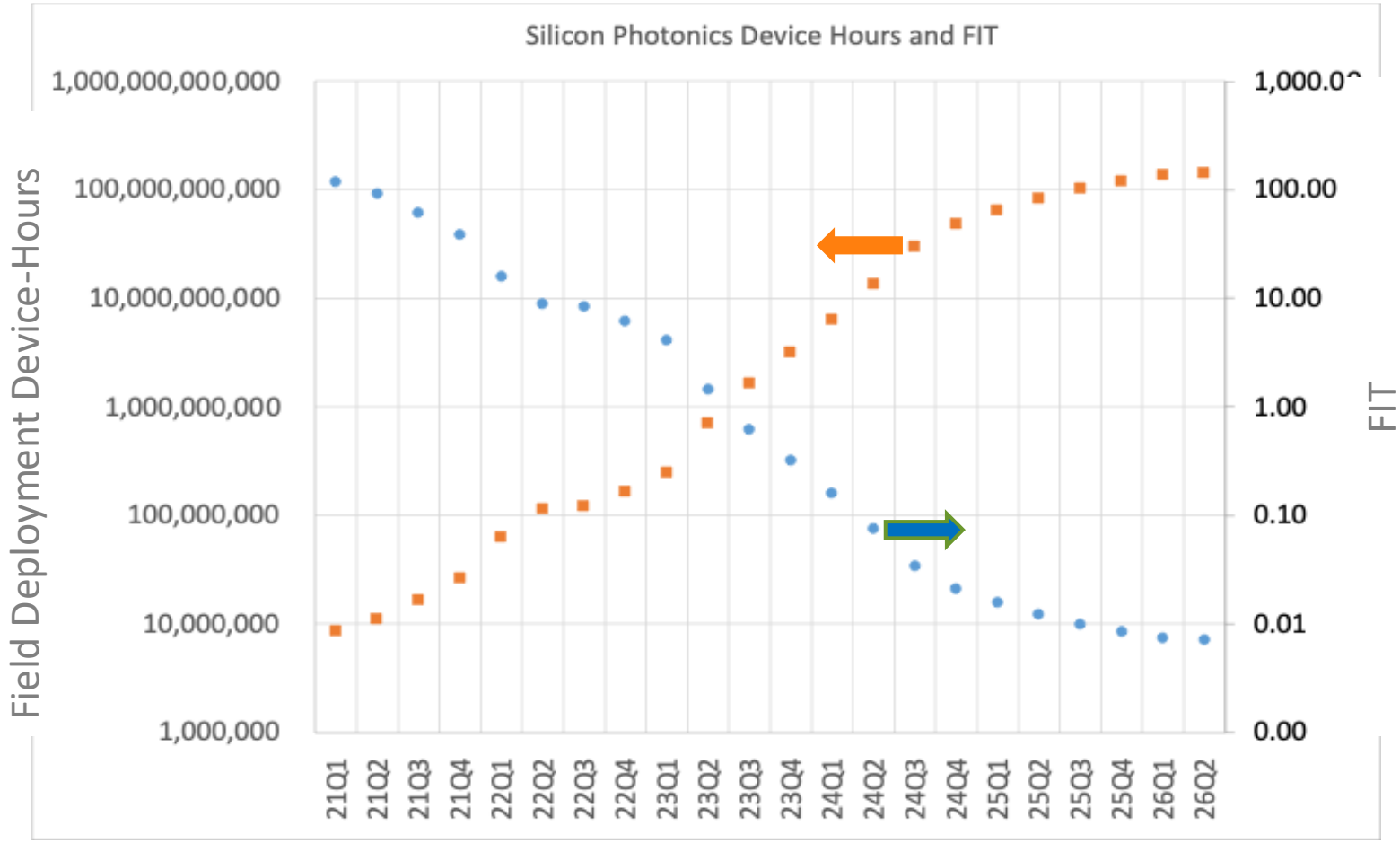


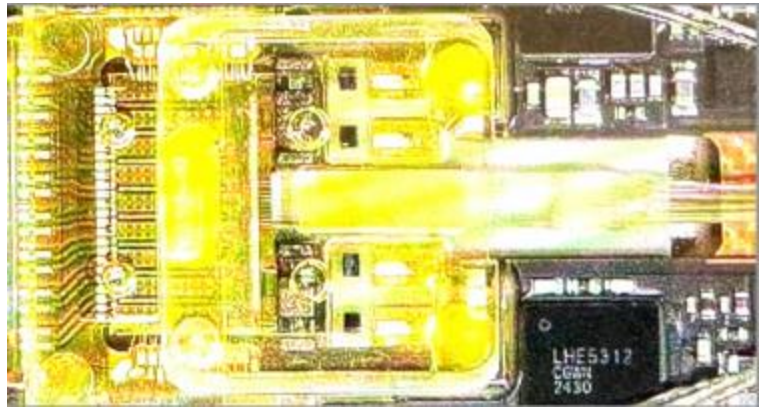
- Dedicated Liquid-Cooling Cold-plate for NPO, keep all critical components cool
- Clean Front Panel with optical I/Os
  - No ELSFPs taking up large front panel space
- Clean SMF fiber bundles to the front panel
  - No extra PM fiber from ELSFPs

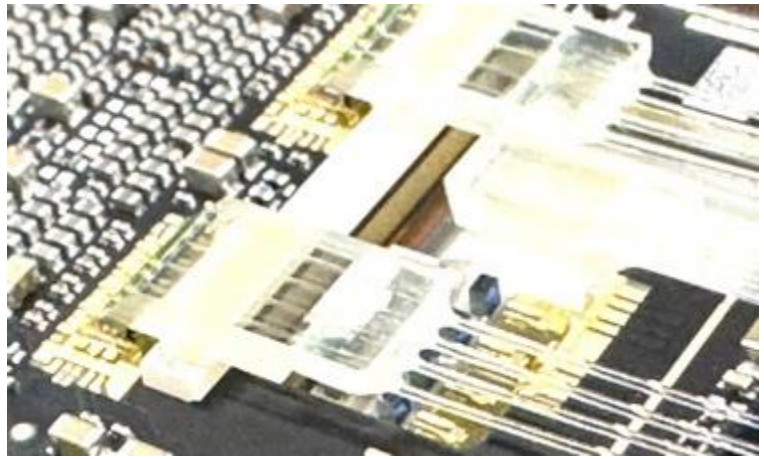
Industry 1st 100Tbps switch Design-Concept by **Arista**, with TeraHop 6.4T "Diablo-1" Open CPX Engine, with permission by Arista

Silicon Photonics PIC Reliability Discussions

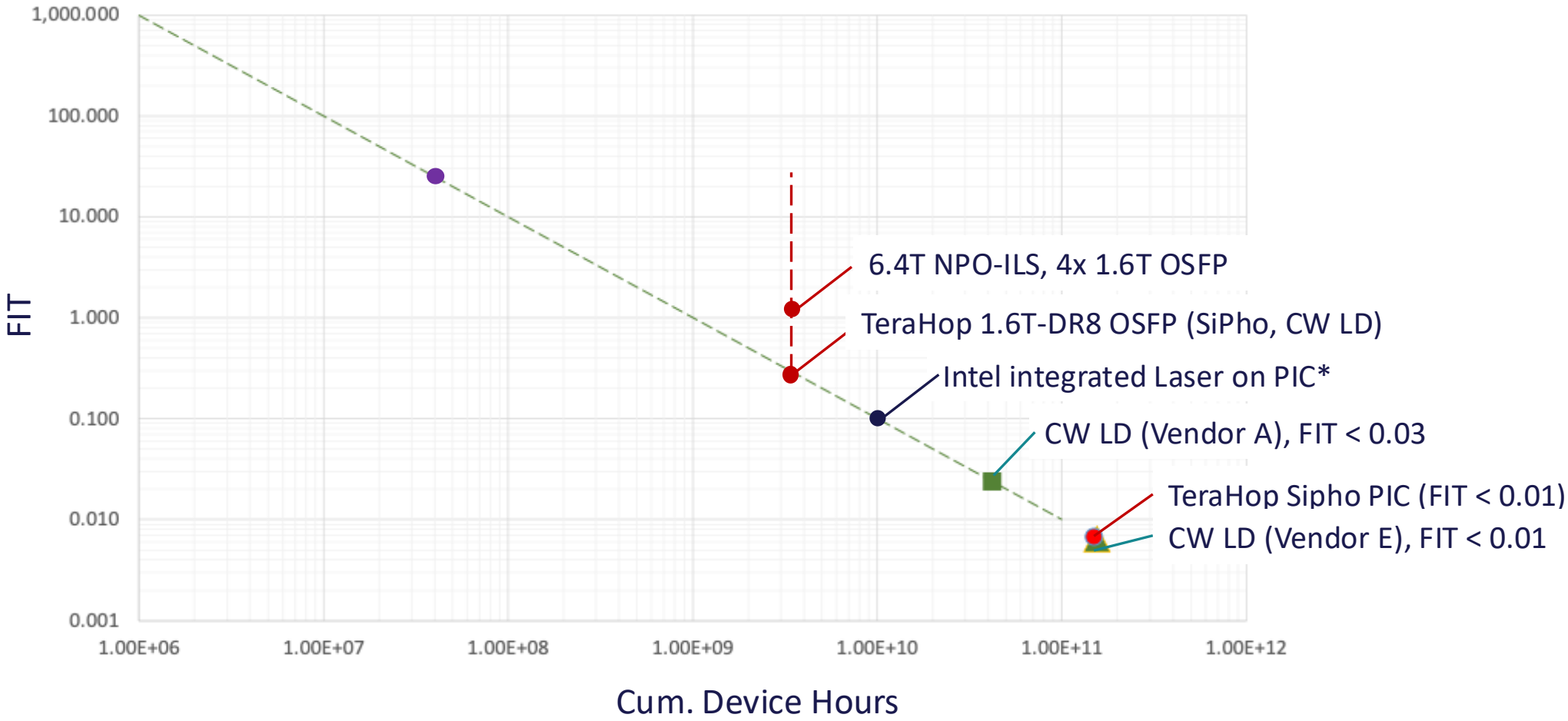
- Not all “silicon photonics” created equal: maturity and high rel. needs field data
- TeraHop Shipped SiPho TRx with PIC >100B Device-Hours, FIT< 0.01







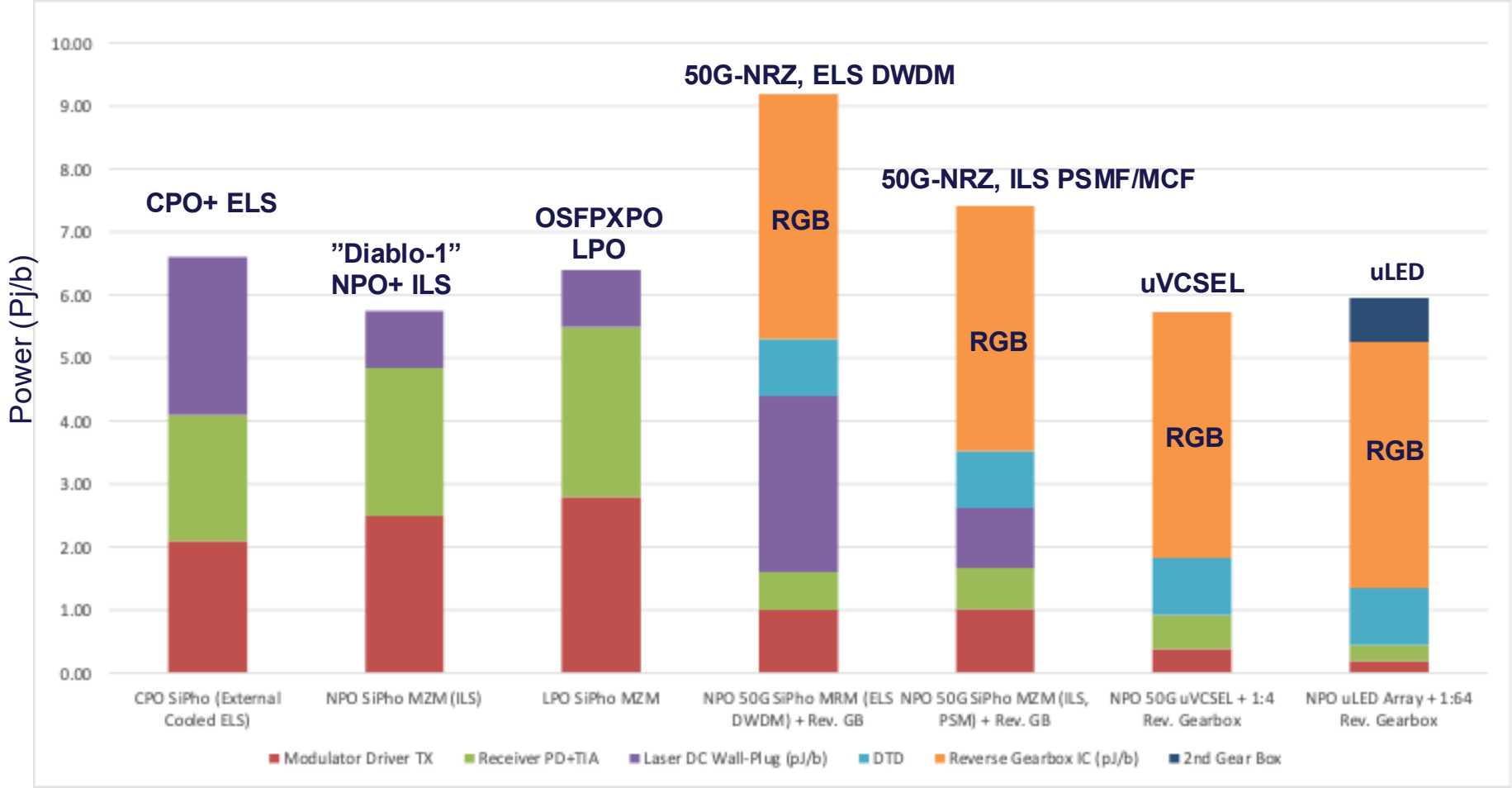
“Diablo-1” Optical Engine Reliability Discussion



* Source: Intel. https://www.intel.com/content/www/us/en/newsroom/news/intel-unveils-first-integrated-optical-io-chiplet.html

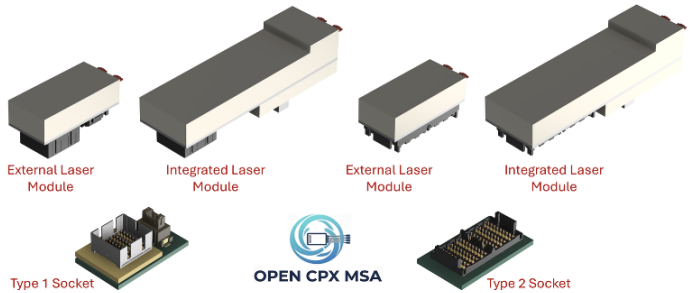
Power Dissipation Discussion

- Mix of copper and optics need to be supported by AI xPUs near term
- “slow and wide” optics requires reverse gear box (RGB) serdes



Open CPX MSA Roadmap

Open CPX MSA Roadmap



Publications

September
Rev 1.0 Spec: 32/36 x 200G
Type 1 & 2, ILM & ELM

March
Rev 1.1 Spec: 32/36 x 200G
Copper media, Type 3, other TBD

June
Rev 2.0 Spec: 64/72 x 200G
Socket type(s), ILM & ELM, Copper

Late 2027 / Early 2028
Rev 3.0 Spec: N x 400G

2026 — 2027 — OFC — 2028

Milestones

March
Open CPX MSA Founded

September
Open CPX Tops 50 Members

OPEN CPX MSA

Subject to change without notice.

Call to Action

- AI infra drive demanding optical interconnect solutions
  - Hi capacity, Hi density, Lower power, High reliability/availability and supply resiliency
- Open optical interconnect solutions keeping up with Innovation
- Please join the Open CPX MSA
  - https://www.opencpxmsa.org/

Thank You!

